

Hide menu

Shor's

Video: Introduction to Shor's

1 min

Video: Shor's Algorithm

22 min

Video: Shor Easy Way

5 min

Video: Quantum Fourier Transform

18 min

Video: Quantum Phase Estimation

24 min

Graded Assignment: Assessment 2

Submitted

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Assessment 2

Next item →

1. In the Deutsch algorithm, **Review Learning Objective** determine if a function is constant or balanced? 1 / 1 point

☐ The amplitude of the superposition state

☐ The state of the second qubit

☐ The phase of the qubits

☒ The state of the first qubit

Help me practice

Let's chat

Correct

Correct! Measuring the state of the first qubit helps determine if the function is constant or balanced.

Assignment details

2. Which component of Qiskit is primarily used for visualizing and simulating quantum algorithms like Grover's search? 1 / 1 point

Due

Dec 18, 11:59 PM IST

Attempts

Unlimited

Submitted

Dec 12, 3:35 AM IST

☐ qiskit-ignis

☐ qiskit-aqua

☒ qiskit-aer

☐ qiskit-terra

Your grade

To pass you need at least 80%. We keep your highest score.

View submission

See feedback

Correct

Correct! Qiskit-aer is used for high-performance quantum computing simulations, including visualization.

Like

Dislike

Report an issue

3. What are some key steps in implementing Grover's search algorithm using Qiskit? 1 / 1 point

☒ Apply the Grover operator multiple times.

Correct

Correct! Repeatedly applying the Grover operator amplifies the probability of measuring the correct state.

☐ Measure the quantum state before applying the Grover operator.

☒ Define the oracle for marking the correct state.

Correct

Correct! Defining the oracle is a crucial step in Grover's search algorithm.

☐ Initialize the qubits in the $|1\rangle$ state.

☒ Create a superposition of all possible states.

Correct

Correct! Creating a superposition is essential for the quantum parallelism in Grover's algorithm.

4. What is the primary significance of Shor's algorithm in quantum computing? 1 / 1 point

☐ It can solve NP-complete problems in polynomial time.

☐ It provides an exponential speedup for searching unsorted databases.

☐ It enables quantum computers to solve differential equations directly.

☒ It can efficiently factorize large numbers, impacting encryption methods.

Correct

Correct! Shor's algorithm can efficiently factorize large numbers, which has significant implications for encryption.

5. Which of the following gates is essential for modular exponentiation in Shor's algorithm? 1 / 1 point

☒ Uf gate

☐ Pauli-X gate

☐ Hadamard gate

☐ CNOT gate

Correct

Correct! The Uf gate is used for modular exponentiation in Shor's algorithm.